Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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Ringkasan

PENDAHULUAN
TINJAUAN PUSTAKA
[12pt]article
[margin=1in]geometry amsmath,amsthm,amssymb,scrextend fancyhdr
Math 310 John Doe 19 Pebruari 2025

Problem x.yz. Delete this text and write theorem statement here. We can draw the sets R, Q, I, Z, and N. Let's assume our problem was: Prove that:

$$(\forall x \in N)\left[\sum_{i=0}^{n} i = \frac{n(n+1)}{2}\right]$$

[1em]0em I will induct on n
Base case (n = 1): $\sum_{i=0}^{1} i = 1 = \frac{1(1+1)}{2} = 1$
Inductive Hypothesis: Assume $\sum_{i=0}^{k} i = \frac{k(k+1)}{2}$ for some $k \in N$
Inductive Step: [I must show: $\sum_{i=0}^{k+1} i = \frac{(k+1)(k+2)}{2}$] $\sum_{i=0}^{k+1} i = k+1+\sum_{i=0}^{k} i[Bydefinitionofseries]$
$= (k+1) + \frac{k(k+1)}{2}[ByI.H]$
$= \frac{(2k+2)+(k^2+k)}{2}$
$= \frac{k^2+3k+2}{3}$
$= \frac{(k+1)(k+1)}{2}$ By the principle of induction, the claim holds for all $n \in N$

Proposition x.yz. Let $n \in Z$.

[1em]0em[Disproof]Blah, blah, blah. I'm so smart.